

Hooks Events Reference

Day 2 Handout - 31 Events & 5 Handler Types (August 2026)

Hooks run external handlers on Claude Code lifecycle events. Use deterministic hooks for formatting, audit logging, and enforceable policy. A prompt or agent handler can add judgment, but it is not a substitute for hard controls.

The 31 Events

1. Session and Instructions (7)

Event	Fires when
SessionStart	A session begins or resumes
Setup	<code>--init-only</code> , or <code>--init</code> / <code>--maintenance</code> in print mode
InstructionsLoaded	CLAUDE.md or rules load into context
ConfigChange	Configuration changes during a session
CwdChanged	The working directory changes
DirectoryAdded	A working directory is added with <code>/add-dir</code> or the SDK
SessionEnd	The session terminates

2. Prompt, Display, and Response (6)

Event	Fires when
UserPromptSubmit	A prompt is submitted, before model processing
UserPromptExpansion	A user-typed command expands into a prompt
MessageDisplay	Assistant text is displayed
Notification	Claude Code sends a notification
Stop	Claude finishes responding
StopFailure	A turn ends because of an API error

3. Tools and Permissions (6)

Event	Fires when
PreToolUse	Before a tool call; can block or modify the decision
PermissionRequest	A tool call requires permission
PermissionDenied	Auto mode denies a tool call
PostToolUse	After a tool call succeeds
PostToolUseFailure	After a tool call fails
PostToolBatch	After a full parallel tool batch resolves

4. Agents and Tasks (5)

Event	Fires when
SubagentStart	A subagent starts
SubagentStop	A subagent finishes
TaskCreated	A tracked task is created
TaskCompleted	A tracked task is marked completed
TeammateIdle	An experimental agent-team teammate is about to idle

5. Workspace and Context (5)

Event	Fires when
FileChanged	A watched file changes on disk
WorktreeCreate	A worktree is about to be created
WorktreeRemove	A worktree is about to be removed
PreCompact	Before context compaction
PostCompact	After context compaction

6. MCP Input (2)

Event	Fires when
Elicitation	An MCP server requests user input
ElicitationResult	The user's response is about to return to the server

The 5 Handler Types

Type	Runs	Best use
command	Local executable or script receiving JSON	Deterministic formatters, blockers, and loggers
http	POST to an endpoint	Central telemetry or policy services
mcp_tool	An MCP tool	Reuse an approved integration
prompt	One model judgment	Bounded fuzzy classification
agent	Multi-step agent with tool access	Verification or investigation requiring several actions

Control rule: a prompt or agent handler adds model judgment. Keep hard security denials in deterministic controls wherever possible.

Example - Format After Edit

```
{
  "hooks": {
    "PostToolUse": [
      {
        "matcher": "Write|Edit",
        "hooks": [
          {
            "type": "command",
            "command": ".claude/hooks/post-edit-format.sh"
          }
        ]
      }
    ]
  }
}
```

Example - Block Before Write

A PreToolUse command handler can return:

```
{
  "hookSpecificOutput": {
    "hookEventName": "PreToolUse",
    "permissionDecision": "deny",
    "permissionDecisionReason": "Hardcoded secret detected"
  }
}
```

Rules That Bite

- 1. Tool matchers are exact and case-sensitive: `Write`, `Edit`, `Bash`.
- 2. Run `/clear` after changing project hook configuration.
- 3. Validate stdout JSON and exit status together.
- 4. Keep high-frequency handlers fast; `/doctor` flags slow hooks.
- 5. Prefer exec form and `${CLAUDE_PROJECT_DIR} / ${CLAUDE_PLUGIN_ROOT}` for paths.
- 6. Treat hook input as untrusted data. Never interpolate it into shell commands without validation.
- 7. Plugin hooks execute with the user's privileges; review the exact source before enabling.

Quick Decisions

Goal	Event	Handler
Block secrets before disk	PreToolUse	command
Format after edits	PostToolUse	command
Record failed tools	PostToolUseFailure	http or command
Verify a completed subagent	SubagentStop	agent
Preserve state before compaction	PreCompact	command or mcp_tool

Goal	Event	Handler
Watch an approved file	FileChanged	command
Apply bounded prompt policy	UserPromptSubmit	prompt

Official reference: <https://code.claude.com/docs/en/hooks>

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